High energy, high luminosity *muon-hadron* collider at CERN

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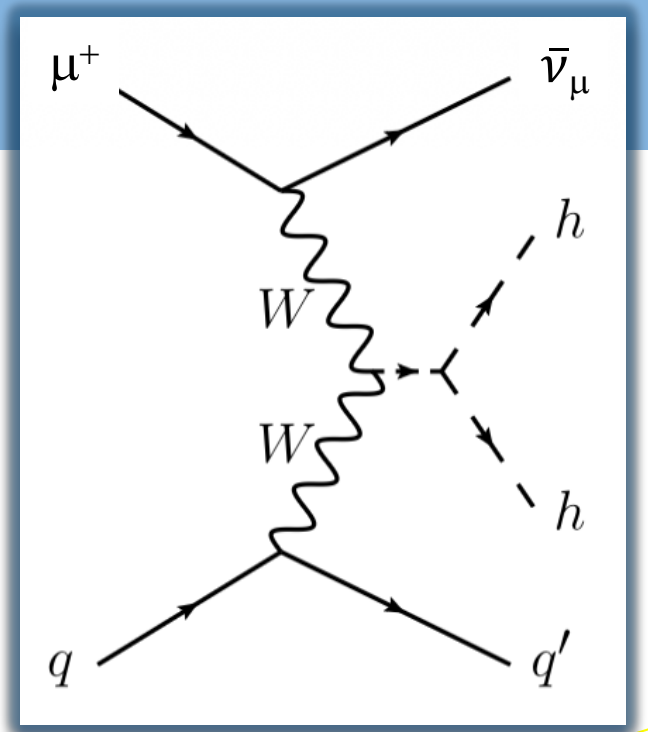
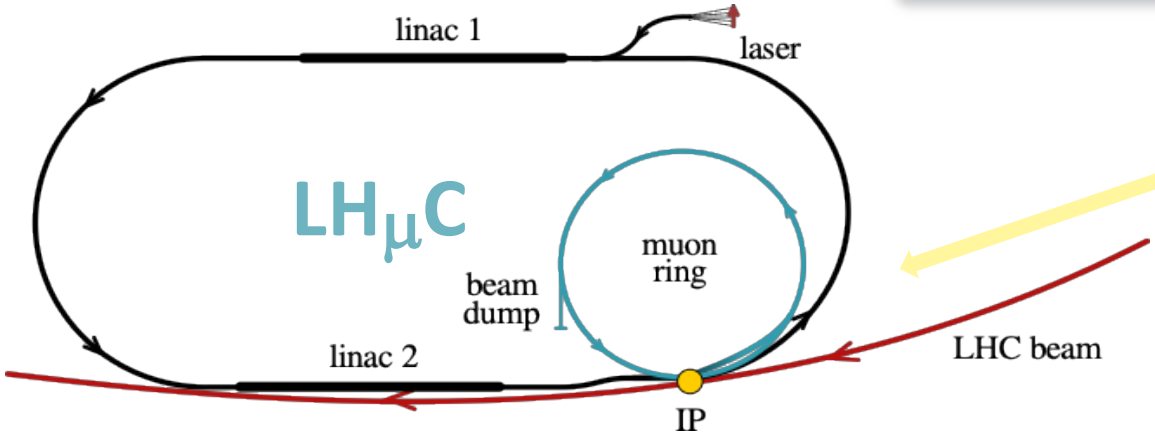
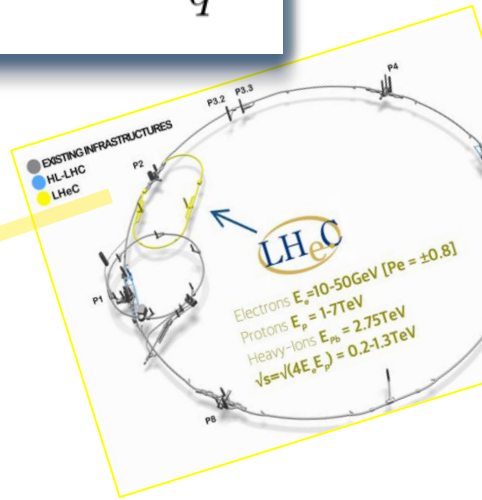
on behalf of HELP4CERN Study Group

DIS 2026



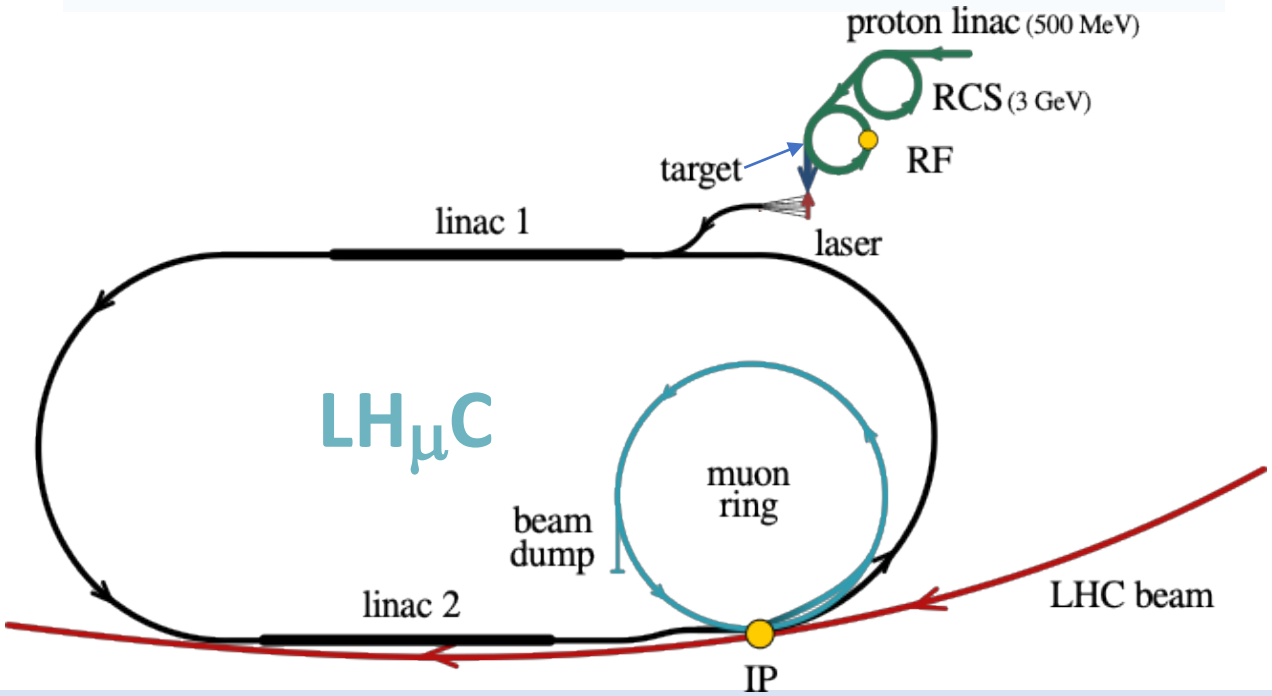




Bringing (anti-)muons to HL-LHC

LHμC proposal for **re-use of LH*e*C infrastructure** to make high energy, high luminosity muon-hadron collisions at HL-LHC (at relatively low incremental cost)

- to add source of ultra-cold anti-muons
- **to convert ERL to booster of anti-muons**
- to add μ^+ storage ring



Nominal μ^+ energy of 1 TeV results in **5.3 TeV** CM energy for muon-proton collisions at HL-LHC with expected luminosity well above **10^{33} cm^{-2}s^{-1}.**

Outstanding scientific scope:

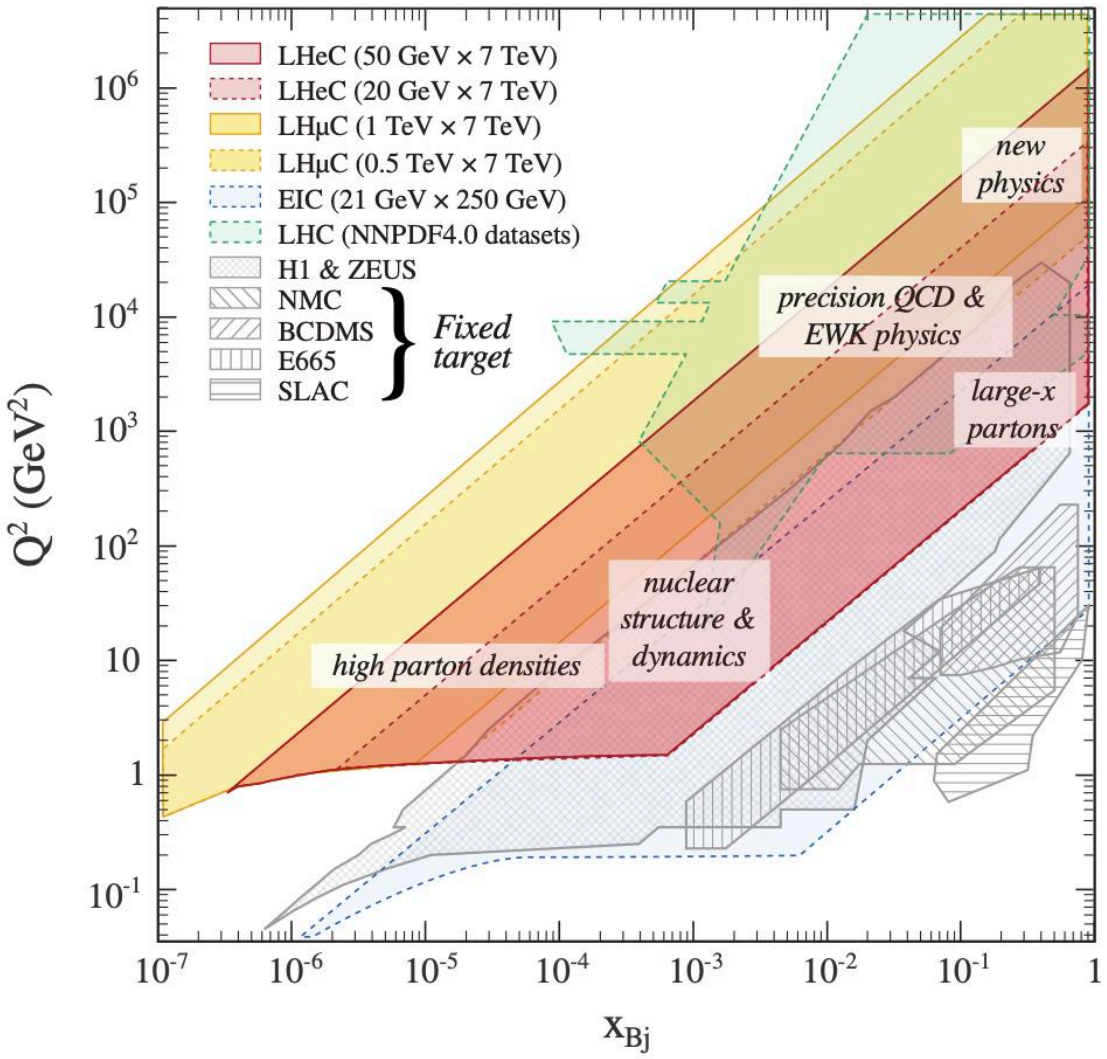
- QCD – unique low-x reach/nuclear effects (and precision hard physics) see talk by P. Kotko
- Impactful contributions to top & electroweak physics
- (Very) exciting prospects for Higgs physics
- Powerful and original BSM searches

see talk by H. Khanpour

Potentially, **large impact for HE-LHC and FCC-*hh*** projects by adding muon-proton collisions at **15 and 30 TeV,** respectively

White Paper in preparation on **H**igh **E**nergy **L**epton-**P**roton collisions at CERN → HELP4CERN Study Group

LHμC: Physics Highlights I



Naturally, LHμC is **very powerful QCD machine:**

see talk by P. Kotko

It extends kinematical reach by order of magnitude wrt LH*e*C, **fully** entering domain of non-linear regime enabling unique and fundamental tests of QCD

Important inclusive measurements can be done **at low luminosity**

Unique precision studies (α_S, fully- & semi-exclusive production and more) – in clean experimental conditions as **event pileup is very low**!

Note: **muon-nucleus collisions** are also of large interest at LHμC

LHμC: Physics Highlights II

LHμC is not just very powerful QCD machine, but thanks to its very high energy and luminosity it is also **powerful HET* Factory**

*) Higgs-Electroweak-Top

see talk by H. Khanpour

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(l\,p \to \nu_l\,H + \text{jet} + \text{X})$ [fb]	52	164	438	731
$\sigma(l\,p \to l\,H + \text{jet} + \text{X})$ [fb]	5.6	19	135	221
$\sigma(l\,p \to l\,t\bar{t} + \text{X})$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(l\,p \to \nu_l\,t + \text{X})$ [pb]	0.55	2.89	21.26	36.77

from HELP4CERN *White Paper* in preparation

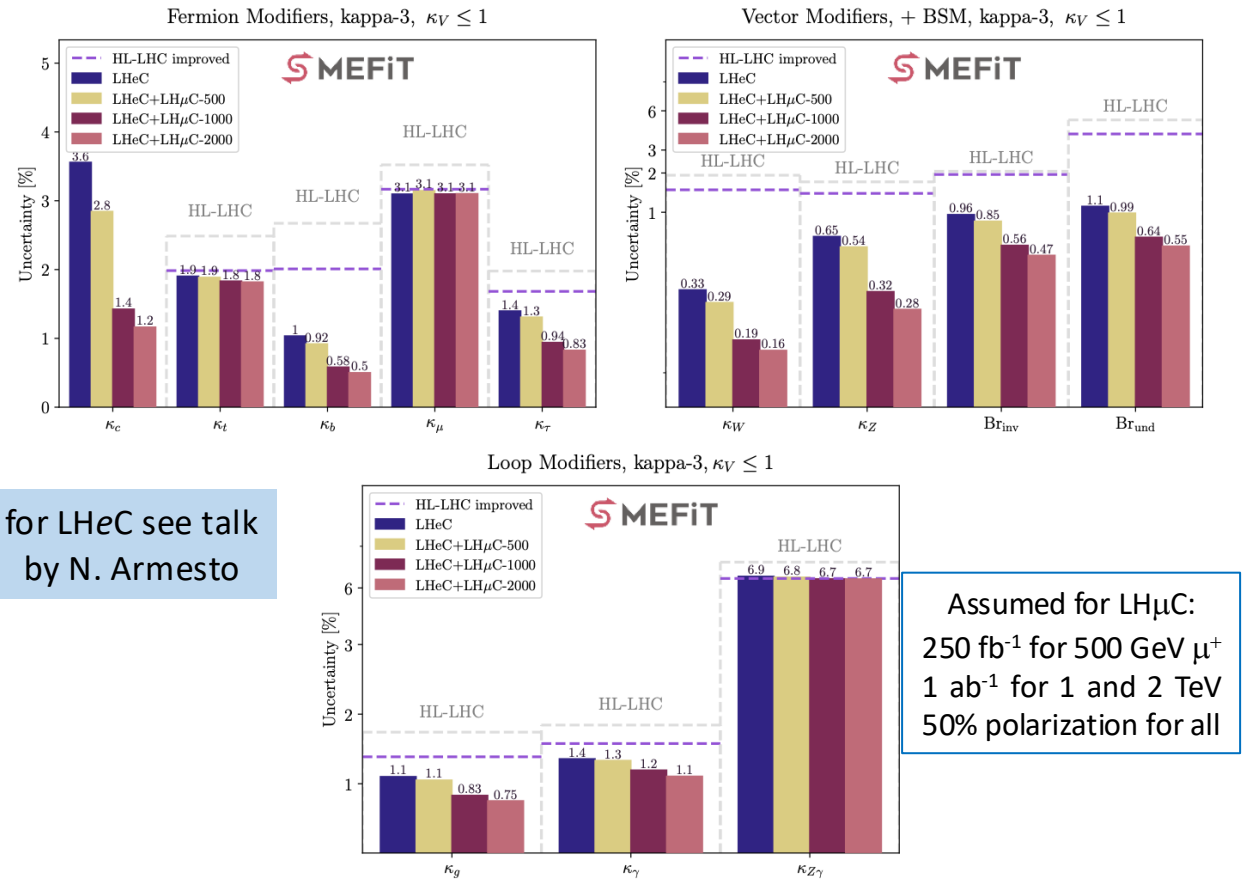
Table 1. Inclusive cross-sections for four major processes at the 20 and 50 GeV LHeC as well as 500 GeV and 1 TeV LHμC. In the LHeC cases an 80% electron beam polarisation is assumed

will continue and enhance rich electroweak program of LH*e*C

+

provide exciting possibilities for **high precision top quark studies** → near-threshold effects and (single-)top couplings

LHμC: Physics Highlights III



for LHeC see talk by N. Armesto

Assumed for LHμC:
250 fb^{-1} for 500 GeV μ$^+$
1 ab^{-1} for 1 and 2 TeV
50% polarization for all

from HELP4CERN *White Paper* in preparation

Figure 2. Relative uncertainty in the coupling modifiers obtained in the kappa-3 framework [58, 59], for different combinations of the HL-LHC, LHeC, and LHμC stages. "HL-LHC improved" refers to the impact of reduced PDF+α_s uncertainties in the HL-LHC measurements as in [4]. The fermion, vector, and loop modifiers are shown from top to bottom respectively. Since neither HL-LHC nor any ℓp collider give direct access to the Higgs width, the condition $\kappa_V(\kappa_W, \kappa_Z) \leq 1$ is imposed on the fit and the uncertainty on these modifiers is defined as $1-\kappa_V(68\%)$. Results have been obtained with the SMEFiT framework [60].

Large improvements are expected in restricting Higgs Yukawa couplings in post-HL-LHC era (significantly surpassing LHeC limits).

Largest sensitivities at LHμC are for *c*- and *b*-quark cases as well as for *W* and *Z* couplings.

Note also impact on limits for invisible and undetected branching fractions.

Thanks to large CM energy, **LHμC offers** exciting possibility of **probing Higgs self-interactions!**

Measurement of double Higgs production under investigation.

Search for R-Parity-Violating (RPV) Supersymmetry at LHμC

Motivation:

- Investigating R-parity-violating (RPV) scenarios within Minimal Supersymmetric Standard Model (MSSM).
- The study focuses on $LQ\overline{D}$-type operators, which enable single resonant production of squarks via s-channel at $\mu^+ p$ colliders.

$$W_{RPV} = \epsilon_i L_i \cdot H_u + \frac{1}{2}\lambda_{ijk} L_i \cdot L_j \bar{E}_k + \lambda'_{ijk} L_i \cdot Q_j \bar{D}_k + \frac{1}{2}\lambda''_{ijk} \bar{U}_i \bar{D}_j \bar{D}_k$$

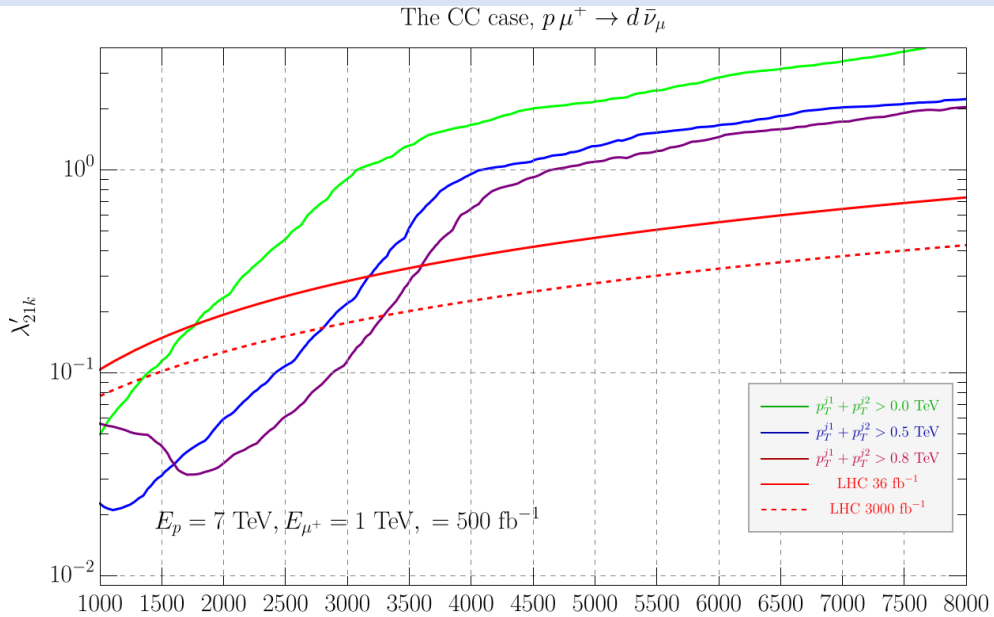
Targeted Channels & Kinematic Strategy:

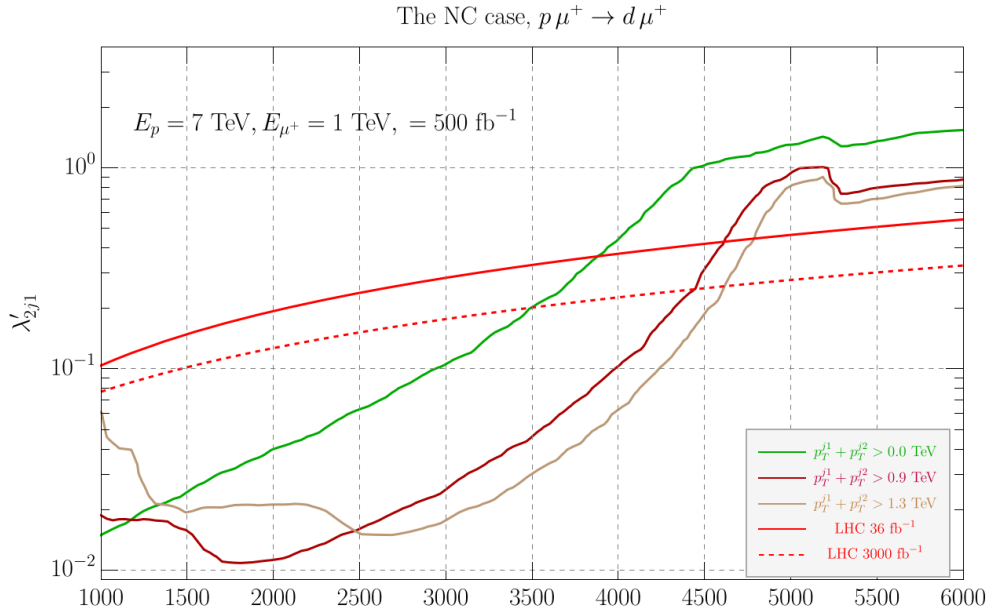
- Charged-Current (CC): $p\,\mu^+ \to \bar{d}\,\bar{\nu}_\mu$ process via a $\tilde{d}^*_R$ mediator.
- Neutral-Current (NC): $p\,\mu^+ \to d\,\mu^+$ process via a $\tilde{u}_L$ mediator.

Signal Optimization: Events are pre-selected with standard parton-level cuts ($p_T^j >$ 5 GeV, $|\eta^j| < 5.5$), requiring exactly 1 muon and at least 1 jet. To isolate signal from background, aggressive jet-p_T-sum thresholds are applied depending on channel.

Key Findings & Discovery Potential (95% C.L.)

- NC channel yields substantially stronger sensitivities than CC channel because NC hard process involves proton valence quarks, whereas CC process relies on sea quarks
- Imposing higher jet-p_T-sum thresholds decisively enhances sensitivity reach for heavier, multi-TeV squarks.
- In NC scenario, proposed collider can test RPV coupling λ'_{2j1} down to 0.01–0.02 for squark masses ranging from 1 to 3 TeV. This capability pushes limits **almost one order of magnitude stronger than current and projected LHC bounds.**



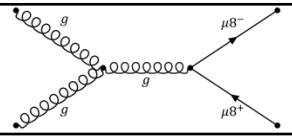
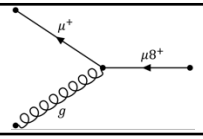


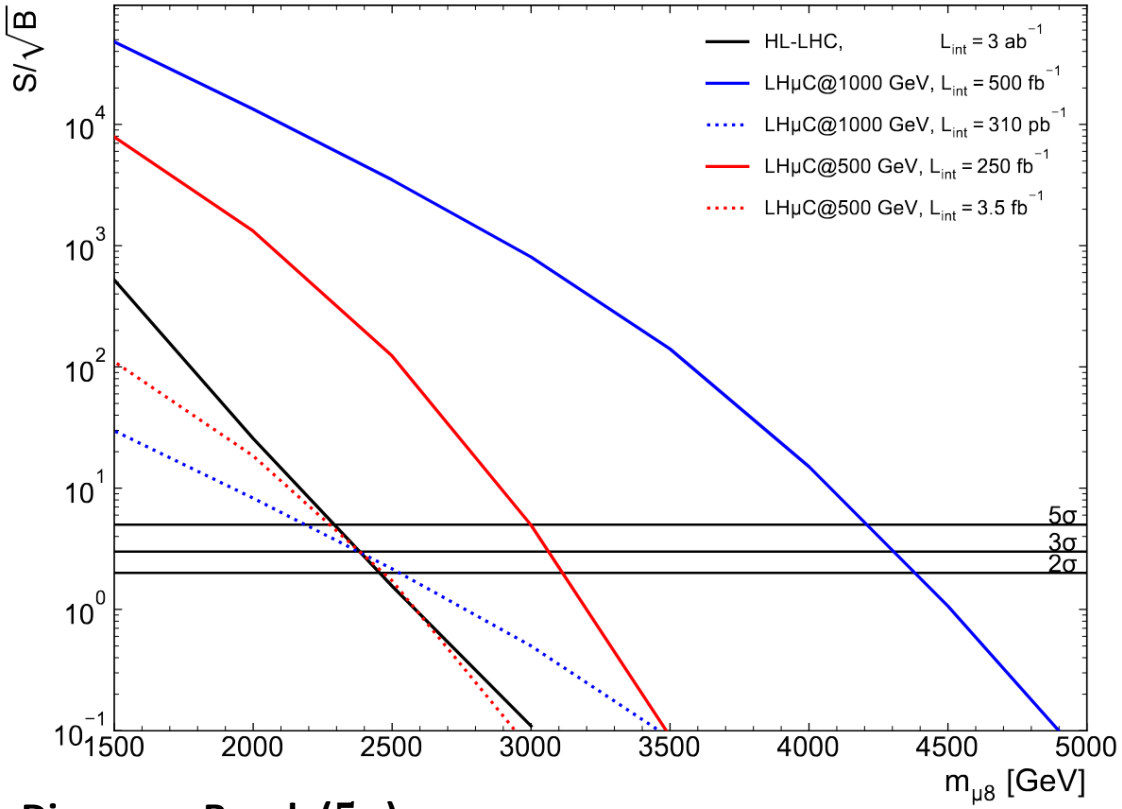
Search for Color-Octet Muons ($\mu 8$) at LHμC

Motivation: In composite models where quarks and leptons are made of fundamental constituents (*preons*), Standard Model muon is accompanied by color-octet partner known as leptogluon ($\mu 8$). This resonance carries QCD color charge while maintaining same electroweak quantum numbers as standard muon, making it prime candidate for probing lepton compositeness.

$$\mathcal{L} = \frac{1}{2\lambda}\{\bar{l}_8^{\alpha} g_s G_{\mu\nu}^{\alpha} \sigma^{\mu\nu}(\eta_L l_L + \eta_R l_R) + h.c.\}$$

Analysis Strategy & Kinematic Cuts:
One exploits near-threshold production of resonance. Table below outlines signal, dominant backgrounds, and specific kinematic cuts applied to optimize signal-to-background ($S/\sqrt{B}$) ratio:

Parameter	HL-LHC (Pair Prod.)	LHμC (Single prod.)
Signal		
Decay	$\mu 8 \to \mu\, j$	
Background	$p\,p \to \mu^+\,\mu^-\,j\,j + X$	$\mu^+\,p \to \mu^+\,j + X$
Required final state	2 muons + 2 jets	1 muon + 1 jet
Leading lepton/jet p_T	> 300 [GeV]	> 400 [GeV]
Sub-leading lepton/jet p_T	> 250 [GeV]	N/A



Discovery Reach (5σ)

- HL-LHC ($1\ ab^{-1}$): ~2.3 TeV
- LHμC@500 GeV ($250\ fb^{-1}$): ~3 TeV
- LHμC@1000 GeV ($500\ fb^{-1}$): ~4.2 TeV

Direct μg initial state at LHμC enables single resonant production, resulting in substantially expanded mass reach. **Notably, LHμC achieves ultimate 3σ observation limit of HL-LHC using remarkably smaller dataset, requiring only 310 pb^{-1} compared to 3 ab^{-1} needed at HL-LHC.**

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When it all began…

PTEP

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μTRISTAN

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The ultra-cold muon technology developed for the muon $g-2$ experiment at J-PARC provides a low-emittance μ^+ beam which can be accelerated and used for realistic collider experiments. We consider the possibility of new collider experiments by accelerating the μ^+ beam up to 1 TeV. Allowing the μ^+ beam to collide with a high-intensity e^- beam at the TRISTAN energy, $E_{e^-} = 30$ GeV, in a storage ring with the same size as TRISTAN (a circumference of 3 km), one can realize a collider experiment with the center-of-mass energy $\sqrt{s} = 346$ GeV, which allows the production of Higgs bosons through vector boson fusion processes. We estimate the deliverable luminosity with existing accelerator technologies to be at the level of $5 \times 10^{33}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$, with which the collider can be a good Higgs boson factory. $\mu^+\mu^+$ colliders up to $\sqrt{s} = 2$ TeV are also possible using the same storage ring. They have the capability of producing the superpartner of the muon up to TeV masses.

PTEP

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μLHC: Antimuon Ring and High Luminosity-LHC-Based μ^+p Collider

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A conceptual design and a performance evaluation of the High Luminosity-Large Hadron Collider (HL-LHC)-based antimuon–proton collider μLHC are presented. Leveraging the μTRISTAN concept based on established Japan Proton Accelerator Research Complex (J-PARC) ultra-cold μ^+ beam technology, μLHC will give the opportunity to achieve a 5.3-TeV center-of-mass energy, significantly surpassing the Electron Ion Collider (EIC) and Large Hadron-electron Collider (LHeC). Two booster ring options for μ^+ acceleration, namely, a μTRISTAN-based and a repurposed LHeC Energy Recovery Linac (ERL)-based system, are explored. Achievable luminosities are predicted to exceed $10^{33}\mathrm{cm}^{-2}\mathrm{s}^{-1}$. The μLHC offers substantially wider kinematic plane coverage, particularly in small-x and high-Q^2 regions, significantly contributing to QCD basics and Higgs boson properties. Its unique potential for Beyond the Standard Model physics extends to muon-related phenomena like excited muons, color-octet muons, leptoquarks, and contact interactions. A possible detector concept is also outlined. Given the maturity of ultra-cold μ^+ beam technology, μLHC is highly feasible for earlier realization than the muon collider, positioning it as a critical tool for the future of high-energy physics.

LHμC: Step 1

Following and further adapting μTRISTAN proposal…

Pion production

Assume 3 GeV RCS similar to that at J-PARC: but 1.6 10^{14} proton bunch and 25 → 50 Hz, 1 → 4 MW

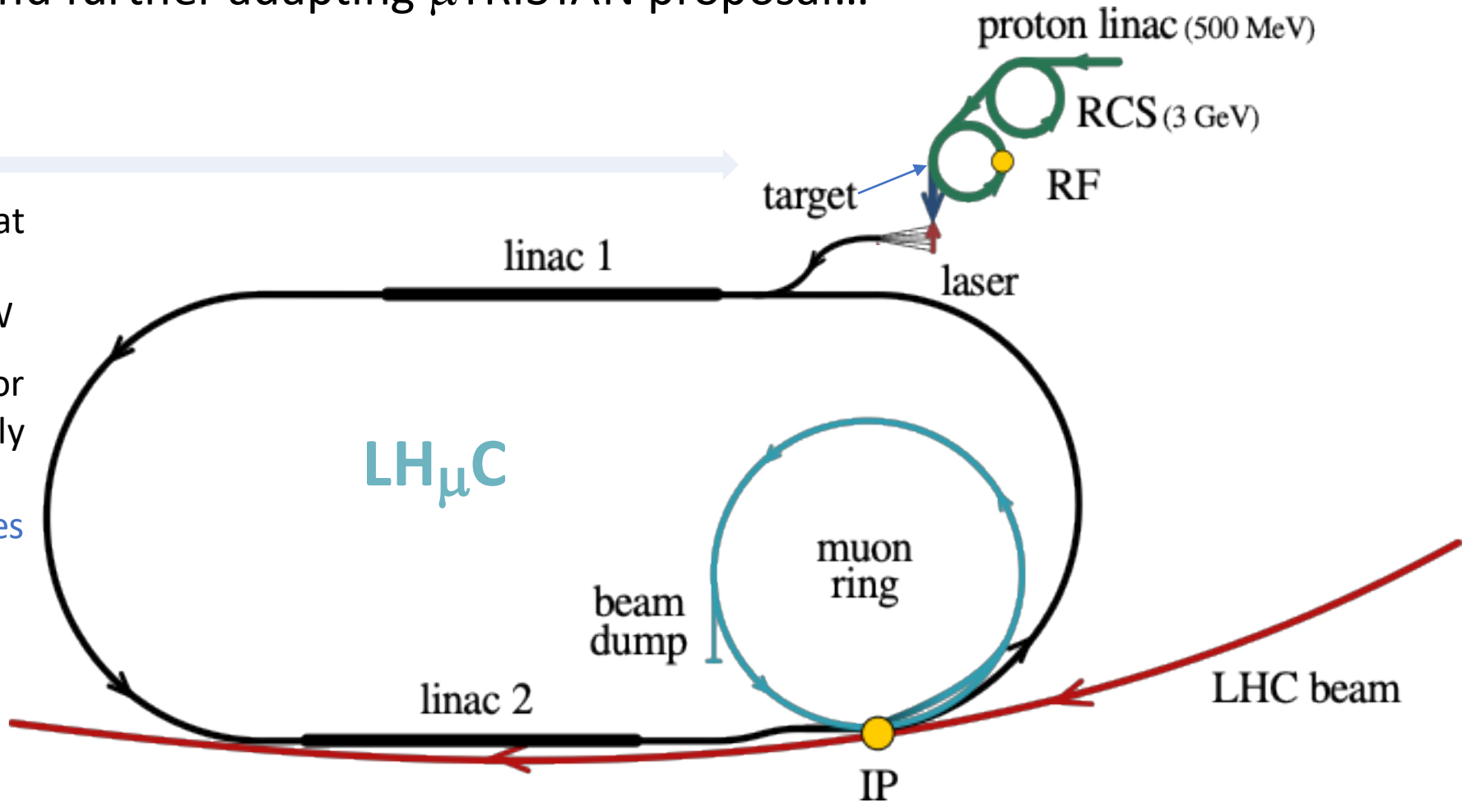
Use another ring with RF cavity for energy compensation to repetitively hit 2 cm thick graphite target

→ produce 40 pion pulses separated by **300 ns**

Target heat load is about 280 kW

Loss of proton beam intensity

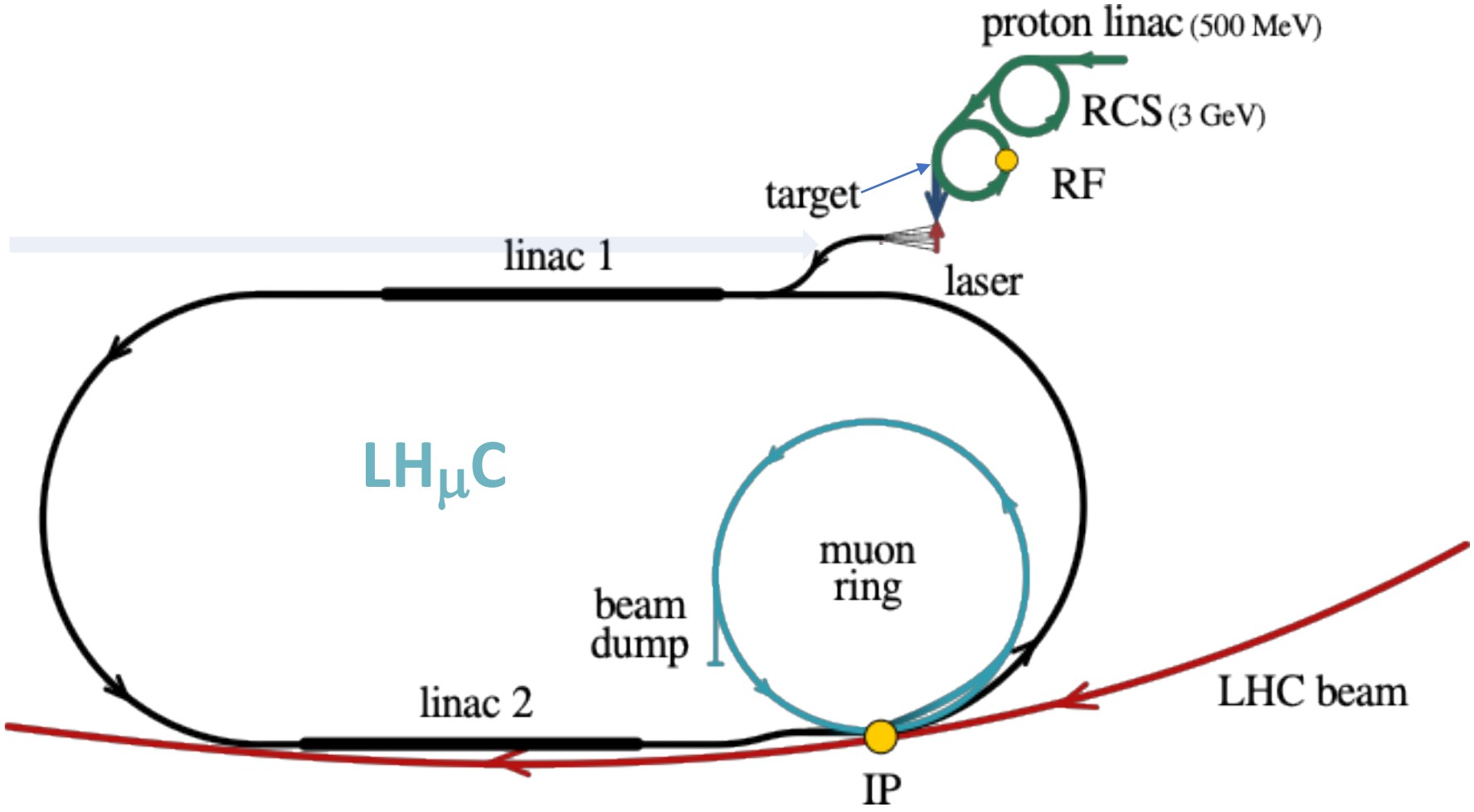
→ ≈20% reduction in pion yield



LHμC: Step 2

Source of ultra-cold μ^+

By use of sophisticated multi-layered scheme – combined, pion stopping, surface muon production and cooling efficiency – increase flux of ultra-cold muons by about **10^6**

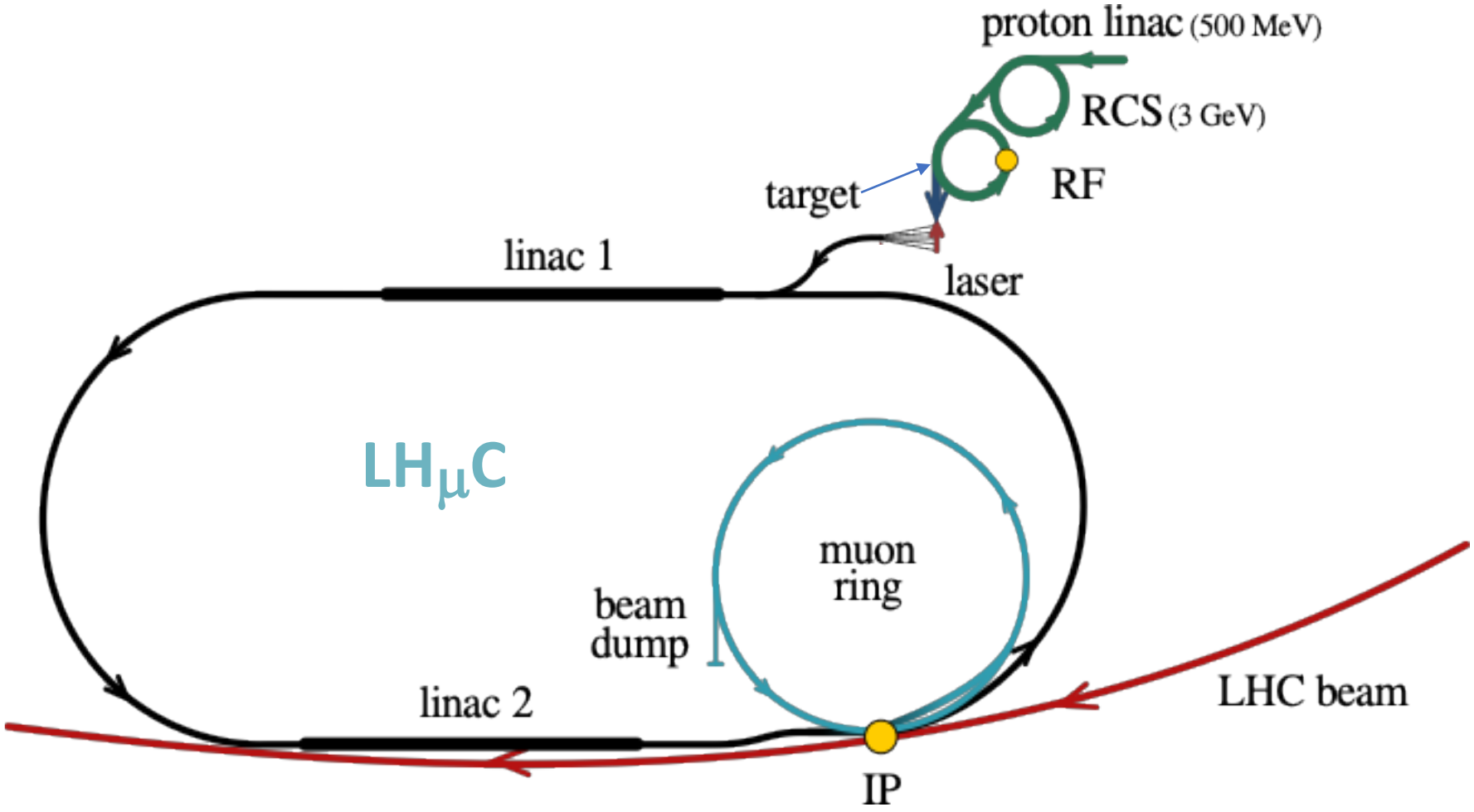


LHμC: Step 3

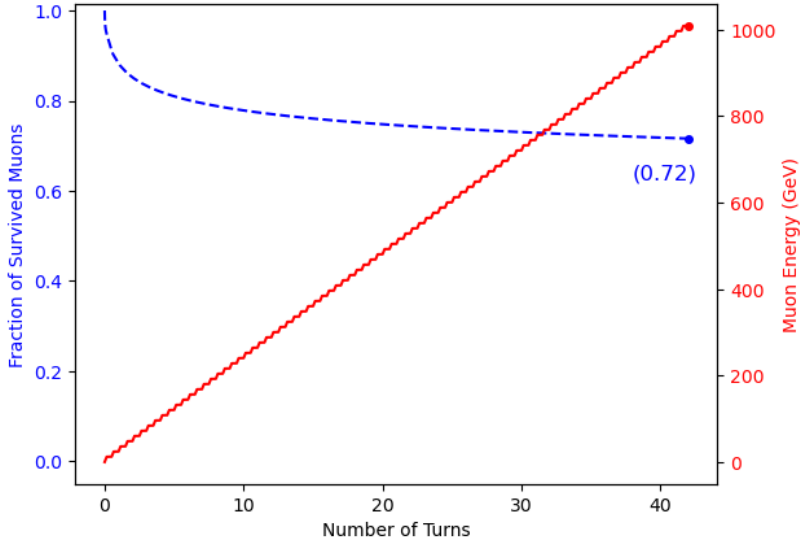
Muon booster

Injection energy = 0.5 GeV, after muon pre-acceleration (under development at J-PARC) → about 20% muons survive.

By **re-use of LHeC linacs** one gets: first, 84.5 GeV muons by **re-using ERL 3-pass arcs**, and then up to 1 TeV by using new vFFA arcs (without RF cavities). In total it requires **42 turns** for muon booster circumference of 8.8 km.



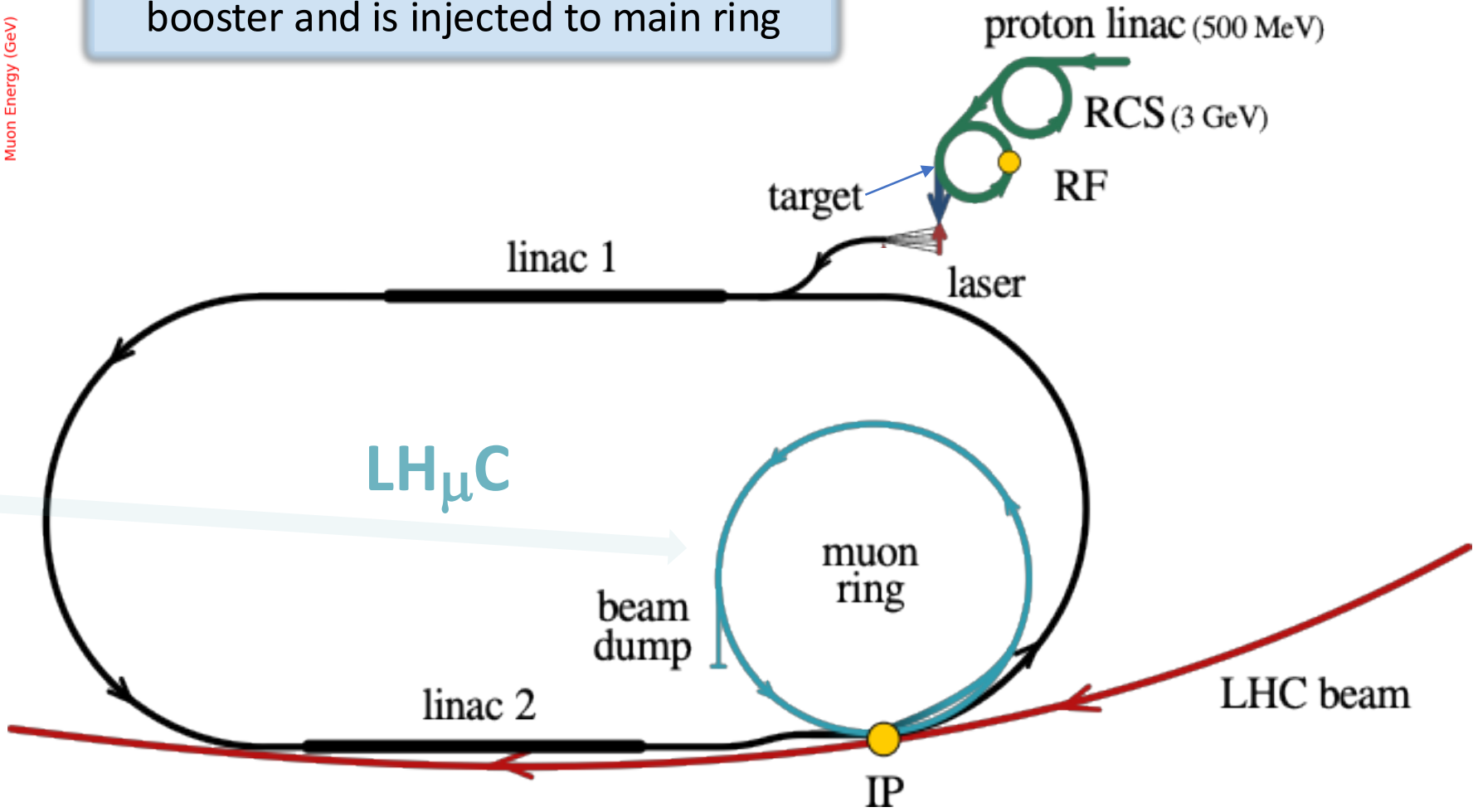
LHμC: Step 4



72% of muons survive acceleration in booster and is injected to main ring

Main muon ring of 3.6 km, its circumference is optimal for 1 TeV top energy – muons are stored for **20 ms** $\simeq \tau_\mu^{LAB}$

Ring of 6.6 km is needed to handle 2 TeV muons to maximize scientific potential – note that extra civil work similar!



LHμC: Performance

Bunch spacing is 300 ns, hence there is possibility of proton bunches twice as populous as nominal ones at HL-LHC, and very small crossing angle is sufficient to avoid parasitic collisions outside IP

Proton beam emittance "fixes" minimal **beam sizes at IP of 4.3 μm**, assuming ultimate HL-LHC optics with $\beta^* = 7$ cm for stand-alone running (no collisions at IP1 and 5)

Assuming 40 bunches entering main ring contain 10^{10} muons each**, peak luminosity of **10^{34} cm^{-2}s^{-1}** is expected for 1 and 2 TeV cases.

Muon polarization of 50%.

) This corresponds to very large initial muon flux at injection to booster of about **3.10^{13} μ/s

Note: Extra power needed to run LHμC is not very large, **below 20 MW**

Antimuons are preferred for integration in LHC (but not quite for some physics)

Neutrino fluxes

LHμC will be very strong source of high energy neutrinos

Two aspects needs addressing/further studies:

- Protection against decays of 1 TeV muons along straight section of main muon ring requires detailed studies, but results for 1.5 TeV case of Muon Collider are very encouraging (500 GeV muon beam should be ”non-problematic” whereas 2 TeV case needs detailed studies)

- **Exciting physics potential thanks to high fluxes of neutrinos** as result of μ^+ decays in Booster!

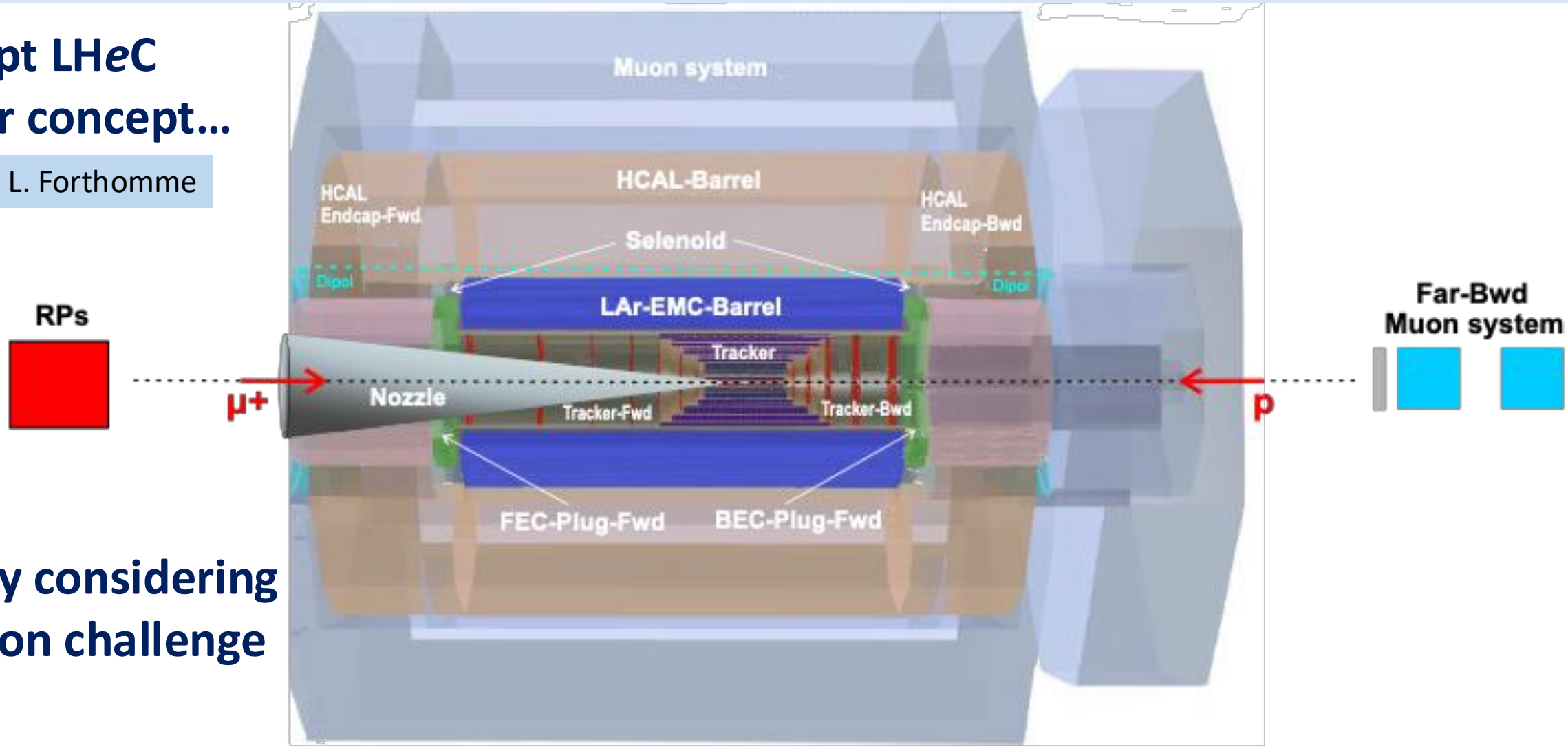
  → ν_e and $\bar{\nu}_\mu$ (w/ bit harder spectrum) in wide energy range from sub-GeV up to hundreds of GeV:

$$\mathbf{d\dot{N}/dE \simeq 3.10^{11}/E}$$ **neutrinos of each type per second per GeV of muon energy *E***

General purpose detector at LHμC

Adapt LH*e*C detector concept…

See talk by L. Forthomme



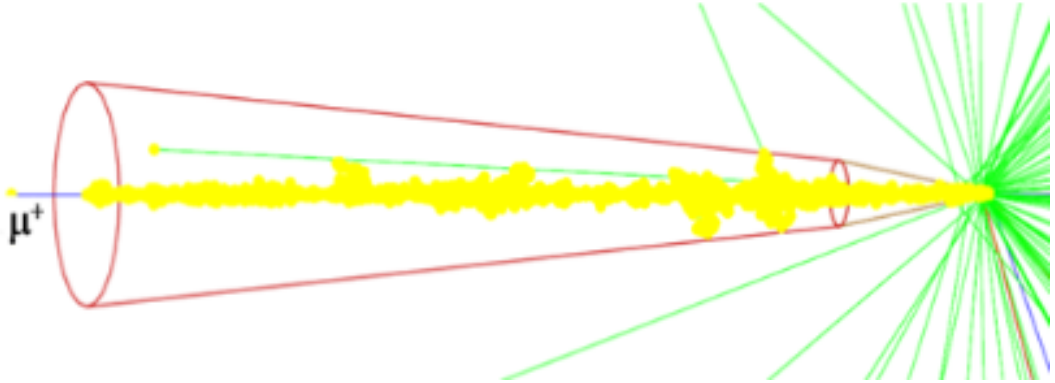
… first by considering TeV muon challenge

Note favorable conditions (as at LH*e*C): **very small event pileup** and possibility of ***triggerless* data streaming**

Detector at LHμC: Muon challenges

- **Need for shielding against beam-induced background**:

  Consider use of conical tungsten nozzle cladded with borated polyethylene, placed around muon beampipe near interaction point – is its half-opening angle of **40**(100) mrad enough, corresponding to $\eta_{max} \simeq$ **4**(3)?



- **Energy resolution of TeV muons:**

  Muon systems at CMS/ATLAS get >**10% energy resolution** for multi-TeV muons – is it **enough at LHμC?** Answer might be different in case of inclusive measurements (which may rely on double-angle, for example) than in case of more exclusive ones. Can resolution be improved by data-driven high statistics re-calibrations?

As outlook: HELP4CERN

White Paper on High Energy, High Luminosity Lepton-Hadron Colliders at CERN

HE*l*-*p*4CERN

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Interested?

Contact: help-4-cern@cern.ch https://help4cern.web.cern.ch

By end of 2026 *White Paper* will provide **comprehensive assessment of v. recent developments** regarding lepton-hadron collisions at LHC, with major two-fold goal in mind:

- propose action plan for developing further LHμC concept/identify its main challenges/suggest next developments
- foster world-wide collaboration/study group towards **preparation by 2028** of proper **"HELP4CERN Proposal"** in close collaboration with LH*e*C Study Group